

IK 514: Introduction to Audio Engineering

Course Code	IK 514
Course Name	Introduction to Audio Engineering
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	BTech/MTech/MS/MSc/MA/Ph.D
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

- **Introduction to Audio Systems:** Overview of various audio systems and components. Signal flow and interfacing standards. (5 Hours)
- **Fundamentals of Sound:** Acoustic principles, sound wave properties, psychoacoustics, and human perception of sound. (10 Hours)
- **Microphones and Speakers:** Types, designs, and applications of microphones and speakers. Practical considerations in microphone placement and speaker setup. (8 Hours)
- **Recording Technology:** Multi-track recording, audio interfaces, digital audio workstations, and recording techniques for different instruments and vocal performances. (8 Hours)
- **Audio Signal Processing:** Equalization, compression, reverb, delay, and other audio effects. Use of audio processing tools in mixing and mastering. (8 Hours)
- **Audio for Video and Film:** Techniques and challenges in audio post-production for video and film, including synchronization, sound design, and Foley. (3 Hours)

Laboratory Modules

1. NA

Text Books

1. Stanley R. Alten, **Audio in Media**, 10th Edition, Cengage Learning, 2014.
2. David Miles Huber, **Modern Recording Techniques**, 9th Edition, Focal Press, 2017.

References

1. Bartlett, B., & Bartlett, J., **Practical Recording Techniques: The Step-by-Step Approach to Professional Audio Recording**, 7th Edition, Focal Press, 2018.
2. Eargle, J., & Foreman, R., **Eargle's The Microphone Book: From Mono to Stereo to Surround - A Guide to Microphone Design and Application**, 3rd Edition, Focal Press, 2020.
3. Izhaki, R., **Mixing Audio: Concepts, Practices, and Tools**, 3rd Edition, Focal Press, 2018.